

CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
OPERATING SYSTEMS (CT104H)
LAB #2

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Chapter 5 :

Question 1: Copy the /etc/services file to the /tmp directory. Open the /tmp/services file in vim, and search for the term WorldWideWeb. Change that to read World Wide Web.

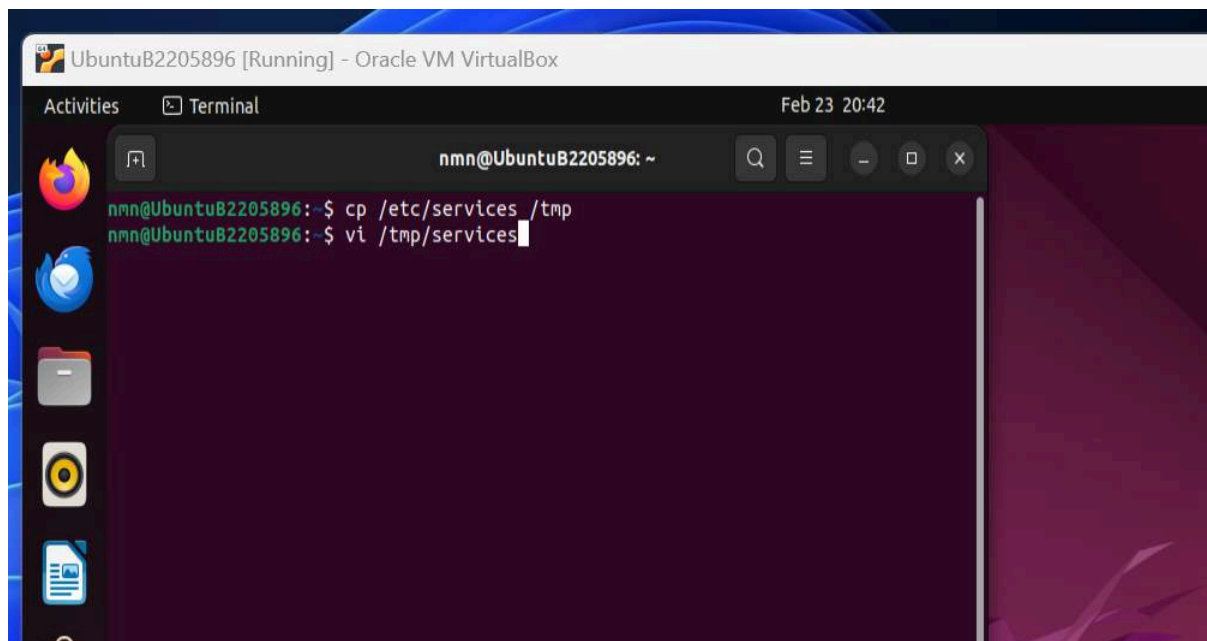
Answer:

```
$ cp /etc/services /tmp
```

```
$ vi /tmp/services
```

```
/WorldWideWeb<Enter>
```

```
cwWorld Wide Web<Esc>
```



```

nmn@UbuntuB2205896: ~
# Network services, Internet style
#
# Updated from https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml .
#
# New ports will be added on request if they have been officially assigned
# by IANA and used in the real-world or are needed by a debian package.
# If you need a huge list of used numbers please install the nmap package.

tcpmux      1/tcp          # TCP port service multiplexer
echo        7/tcp
echo        7/udp
discard     9/tcp          sink null
discard     9/udp          sink null
sysstat     11/tcp         users
daytime     13/tcp
daytime     13/udp
netstat     15/tcp
qotd        17/tcp          quote
chargen     19/tcp          ttytst source
chargen     19/udp          ttytst source
ftp-data    20/tcp
ftp         21/tcp
/WorldWideWeb

```

```

nmn@UbuntuB2205896: ~
time        37/udp          tinserver
whois       43/tcp          nicname
tacacs      49/tcp          # Login Host Protocol (TACACS)
tacacs      49/udp
domain      53/tcp          # Domain Name Server
domain      53/udp
bootps      67/udp
bootpc      68/udp
tftp        69/udp
gopher      70/tcp          # Internet Gopher
finger      79/tcp
http        80/tcp          # WorldWideWeb HTTP
kerberos    88/tcp          kerberos5 krb5 kerberos-sec # Kerberos v5
kerberos    88/udp          kerberos5 krb5 kerberos-sec # Kerberos v5
iso-tsap    102/tcp         # part of ISODE
acr-nema    104/tcp         # Digital Imag. & Comm. 300
pop3        110/tcp         # POP version 3
sunrpc      111/tcp         # RPC 4.0 portmapper
sunrpc      111/udp         portmapper
auth        113/tcp         authentication tap ident
nntp        119/tcp         readnews untp # USENET News Transfer Protocol
ntp         123/udp         # Network Time Protocol
epmap       135/tcp         loc-srv       # DCE endpoint resolution

```

```

nmn@UbuntuB2205896: ~
time 37/udp timserver
whois 43/tcp nicname
tacacs 49/tcp # Login Host Protocol (TACACS)
tacacs 49/udp
domain 53/tcp # Domain Name Server
domain 53/udp
bootps 67/udp
bootpc 68/udp
tftp 69/udp
gopher 70/tcp # Internet Gopher
finger 79/tcp
http 80/tcp www # World Wide Web HTTP
kerberos 88/tcp kerberos5 krb5 kerberos-sec # Kerberos v5
kerberos 88/udp kerberos5 krb5 kerberos-sec # Kerberos v5
iso-tsap 102/tcp tsap # part of ISODE
acr-nema 104/tcp dicom # Digital Imag. & Comm. 300
pop3 110/tcp pop-3 # POP version 3
sunrpc 111/tcp portmapper # RPC 4.0 portmapper
sunrpc 111/udp portmapper
auth 113/tcp authentication tap ident
nntp 119/tcp readnews untp # USENET News Transfer Protocol
ntp 123/udp # Network Time Protocol
epmap 135/tcp loc-srv # DCE endpoint resolution
  
```

Question 2: Find the following paragraph in your /tmp/services file (if it is not there, choose a different paragraph) , and move it to the end of that file.

Answer:

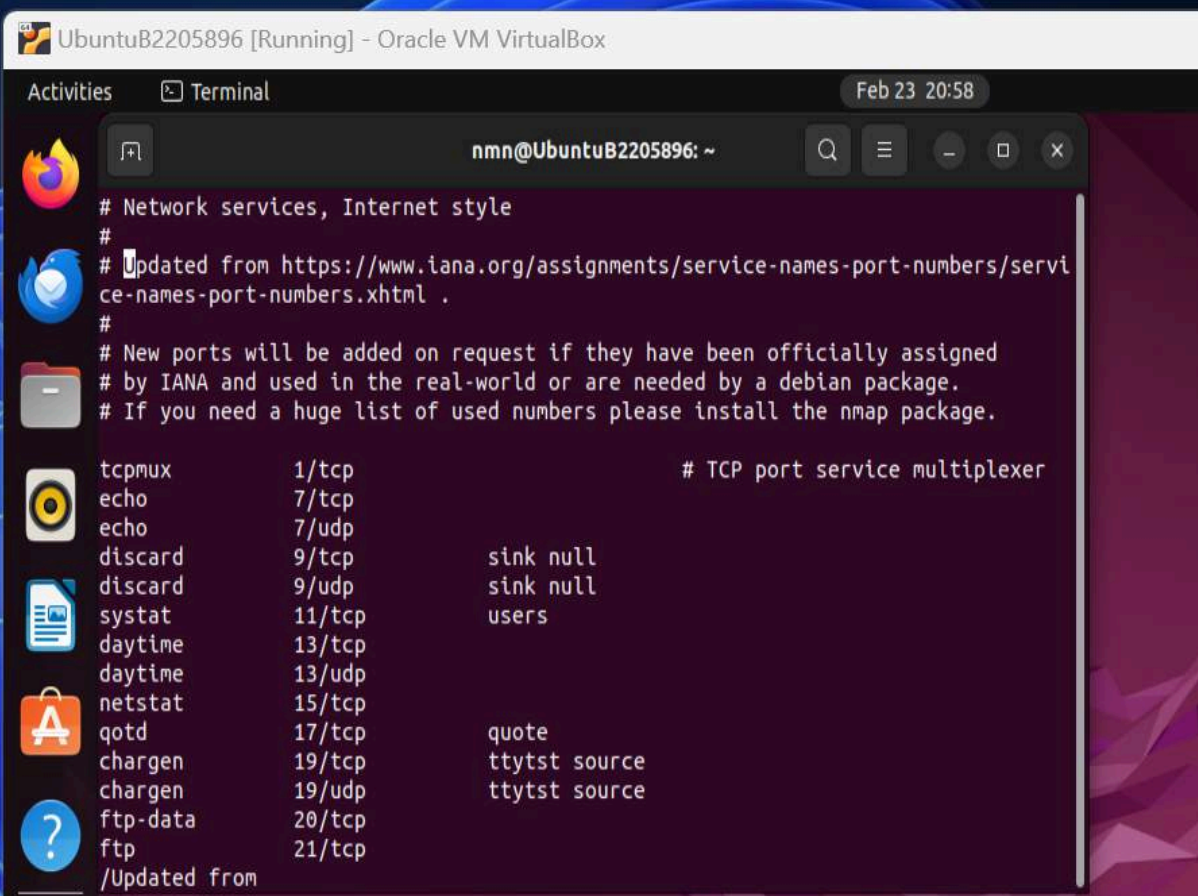
\$ vi /tmp/services

/Updated from<Enter>

4dd

G

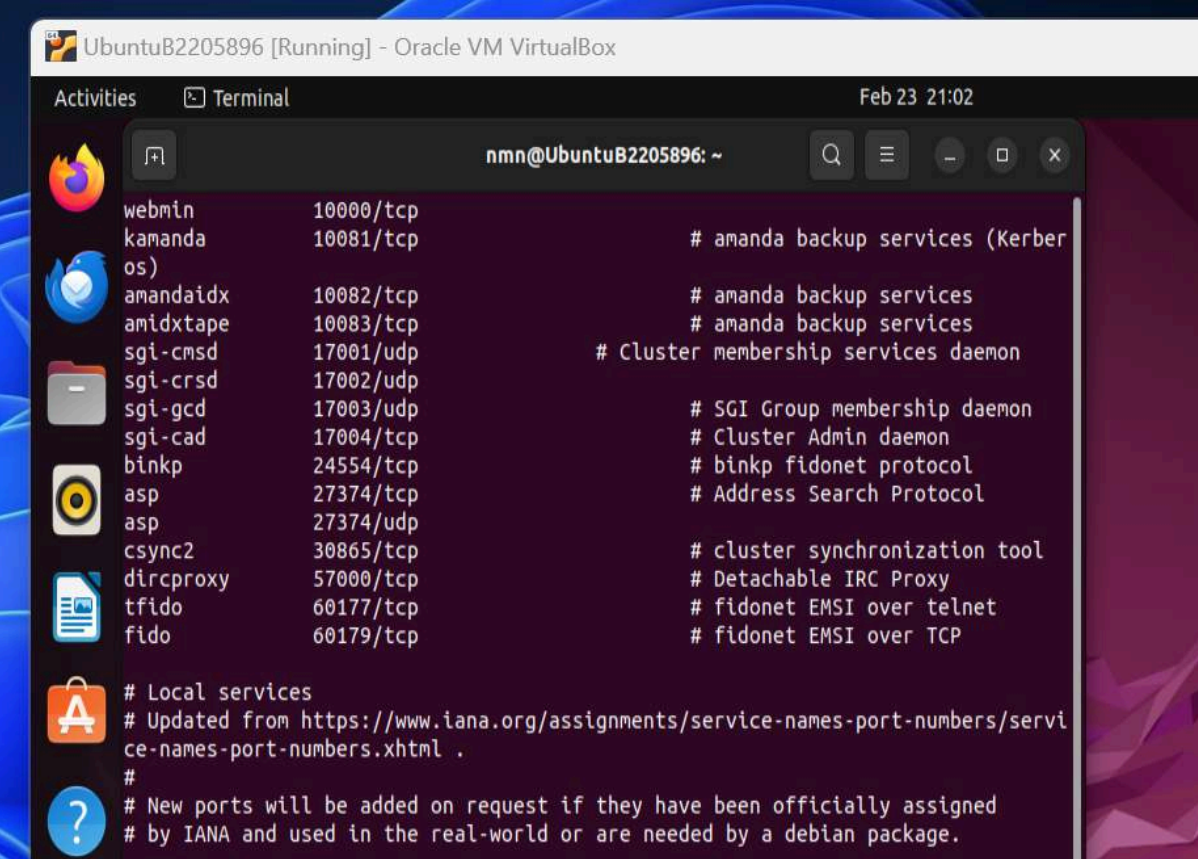
p



Terminal window showing network services configuration in Ubuntu. The window title is "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The terminal output includes a list of network services and their corresponding ports, along with comments about IANA assignments and the nmap package.

```
nmn@UbuntuB2205896: ~
# Network services, Internet style
#
# Updated from https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml .
#
# New ports will be added on request if they have been officially assigned
# by IANA and used in the real-world or are needed by a debian package.
# If you need a huge list of used numbers please install the nmap package.

tcpmux          1/tcp          # TCP port service multiplexer
echo            7/tcp
echo            7/udp
discard         9/tcp          sink null
discard         9/udp          sink null
sysstat         11/tcp         users
daytime         13/tcp
daytime         13/udp
netstat         15/tcp
qotd            17/tcp         quote
chargen         19/tcp         ttytst source
chargen         19/udp         ttytst source
ftp-data        20/tcp
ftp             21/tcp
/Updated from
```



Terminal window showing local services configuration in Ubuntu. The window title is "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The terminal output includes a list of local services and their corresponding ports, along with comments about IANA assignments and the nmap package.

```
nmn@UbuntuB2205896: ~
webmin          10000/tcp
kamanda         10081/tcp      # amanda backup services (Kerber
os)
amandaidx       10082/tcp      # amanda backup services
amidxtape       10083/tcp      # amanda backup services
sgi-cmsd        17001/udp      # Cluster membership services daemon
sgi-crsd        17002/udp
sgi-gcd         17003/udp      # SGI Group membership daemon
sgi-cad         17004/tcp      # Cluster Admin daemon
binkp           24554/tcp      # binkp fidonet protocol
asp             27374/tcp      # Address Search Protocol
asp             27374/udp
csync2          30865/tcp      # cluster synchronization tool
dircproxy       57000/tcp      # Detachable IRC Proxy
tfido           60177/tcp      # fidonet EMSI over telnet
fido            60179/tcp      # fidonet EMSI over TCP

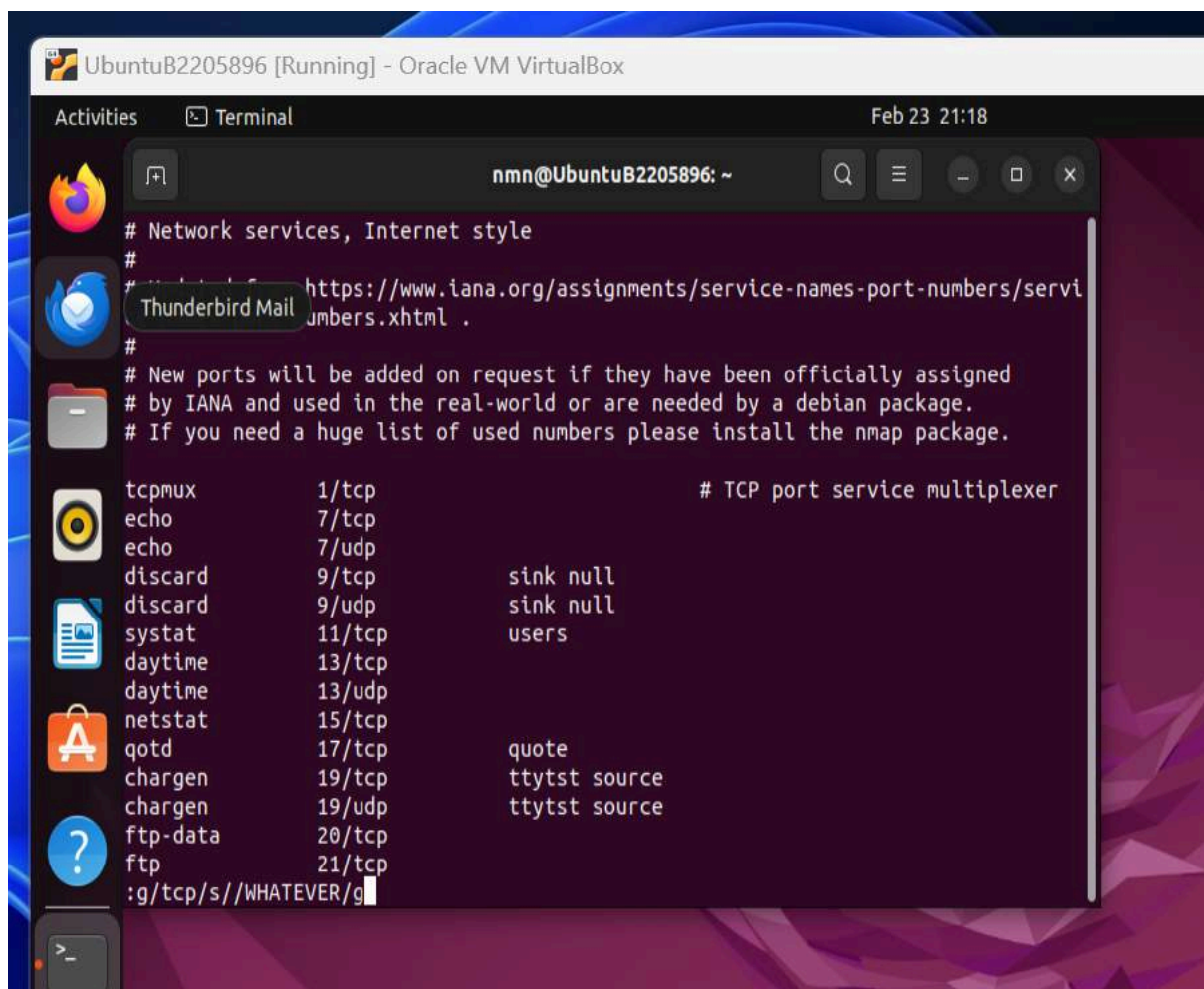
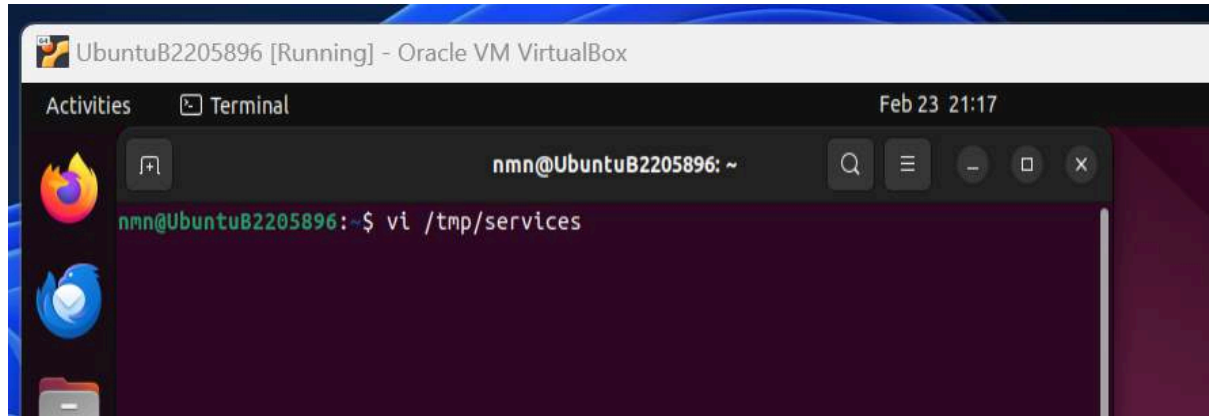
# Local services
# Updated from https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml .
#
# New ports will be added on request if they have been officially assigned
# by IANA and used in the real-world or are needed by a debian package.
```

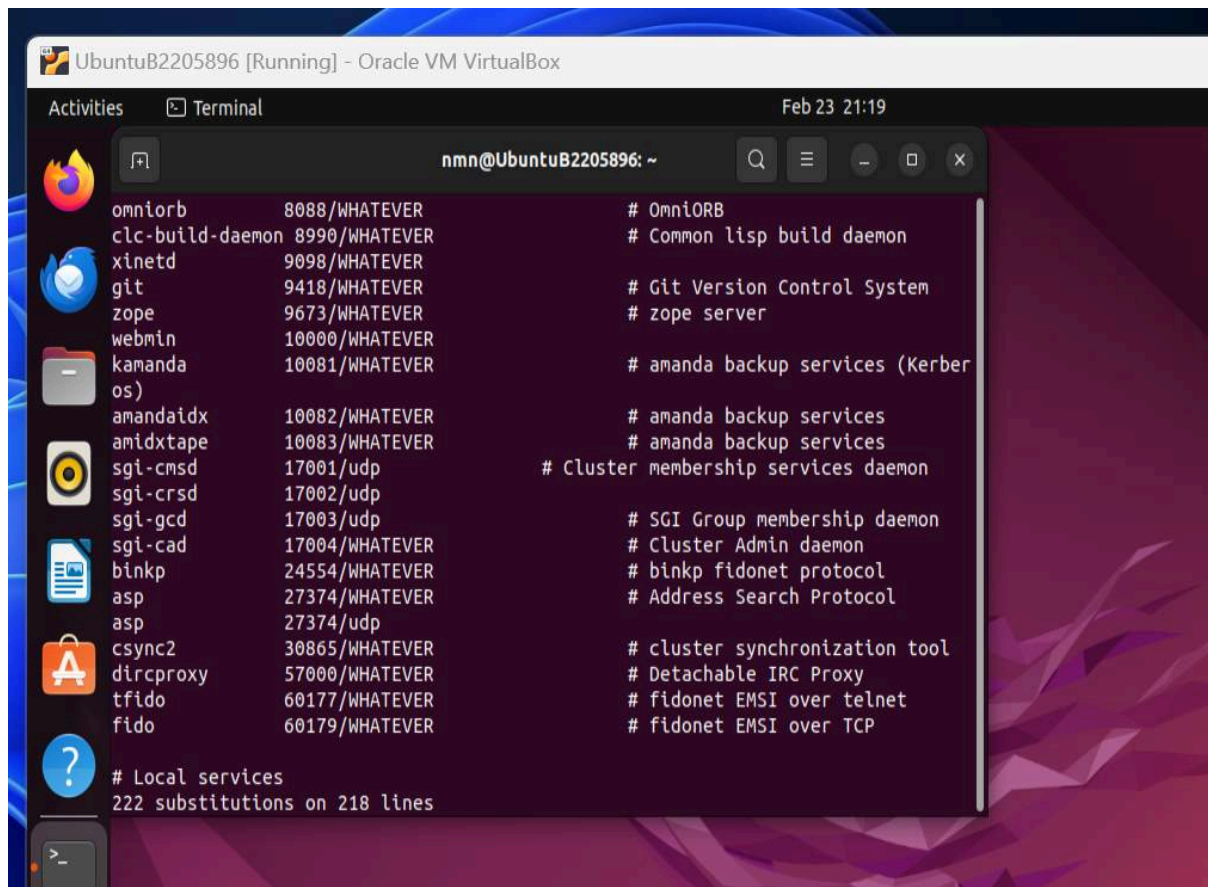
Question 3: Using ex mode, search for every occurrence of the term tcp (case-sensitive) in your /tmp/services file and change it to WHATEVER.

Answer:

\$ vi /tmp/services

:g/tcp/s//WHATEVER/g<Enter>

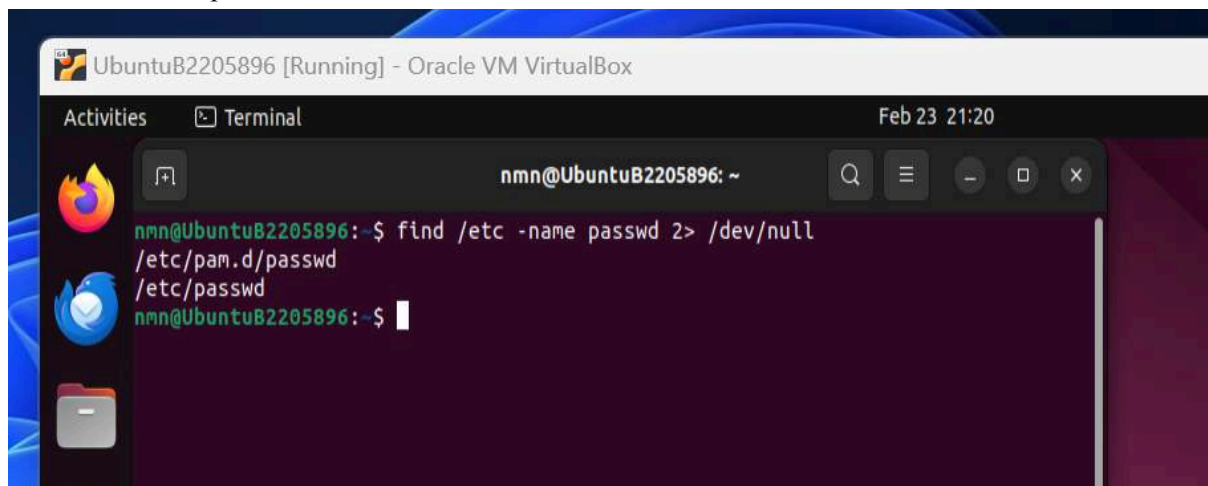




Question 4: As a regular user, search the /etc directory for every file named passwd. Redirect error messages from your search to /dev/null.

Answer:

```
$ find /etc -name passwd 2> /dev/null
```



Question 5: Create a directory in your home directory called TEST. Create files in that directory named one, two, and three that have full read/write/execute permissions on for everyone (user, group, and other). Construct a find command to find those files and any other files that have write permission open to “others” from your home directory and below.

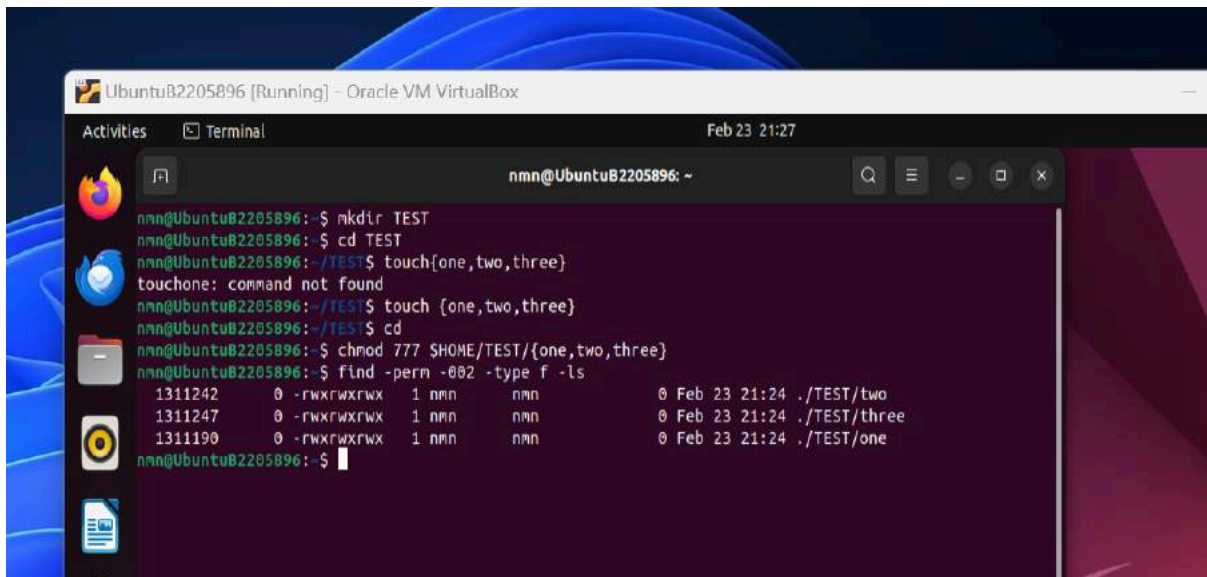
Answer:

```
$ mkdir TEST
```

```
$ cd TEST
```

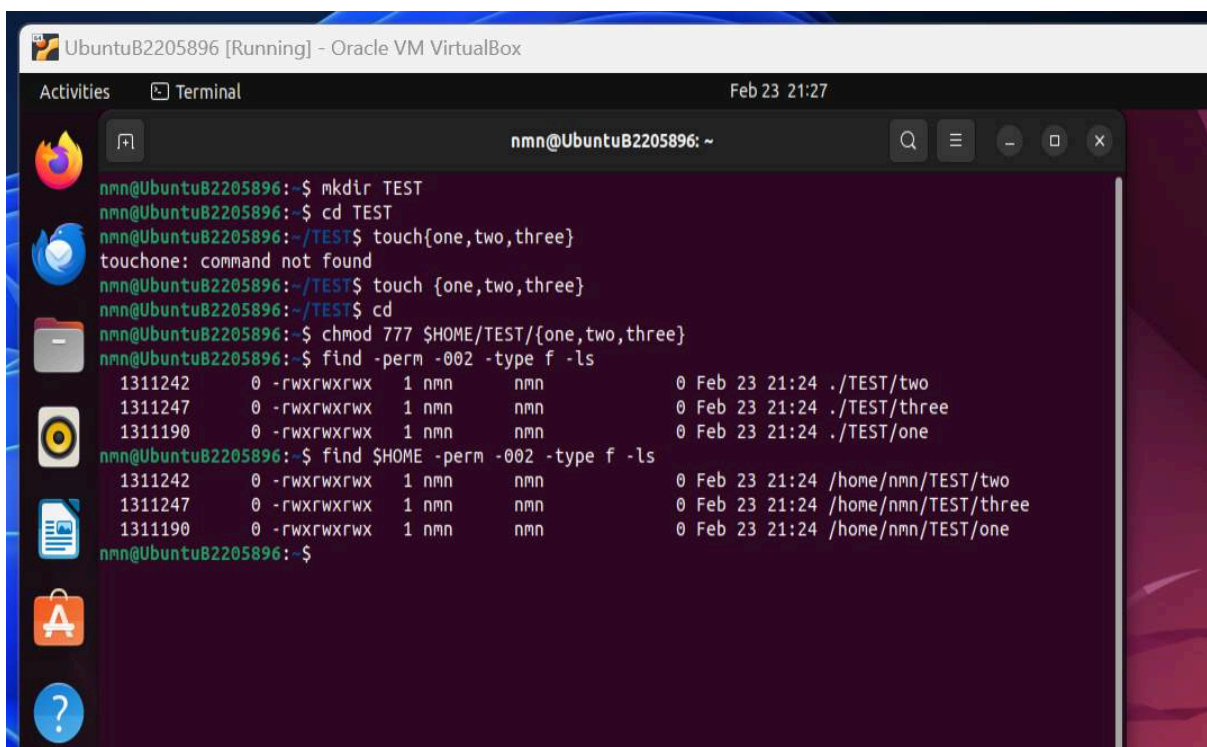


```
$ touch {one,two,three}
$ cd
$ chmod 777 $HOME/TEST/{one,two,three}
$ find -perm -002 -type f -ls
```



A terminal window titled 'UbuntuB2205896 [Running] - Oracle VM VirtualBox' showing a user named 'nmn' performing several commands. The user creates a directory 'TEST', changes to it, and uses 'touch' to create files 'one', 'two', and 'three'. Then, 'chmod 777' is used to set permissions on these files. Finally, 'find -perm -002 -type f -ls' is executed, listing the files with their permissions, ownership, and modification times.

```
nmn@UbuntuB2205896:~$ mkdir TEST
nmn@UbuntuB2205896:~$ cd TEST
nmn@UbuntuB2205896:~/TEST$ touch{one,two,three}
touchone: command not found
nmn@UbuntuB2205896:~/TEST$ touch {one,two,three}
nmn@UbuntuB2205896:~/TEST$ cd
nmn@UbuntuB2205896:~$ chmod 777 $HOME/TEST/{one,two,three}
nmn@UbuntuB2205896:~$ find -perm -002 -type f -ls
1311242 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 ./TEST/two
1311247 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 ./TEST/three
1311190 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 ./TEST/one
nmn@UbuntuB2205896:~$
```



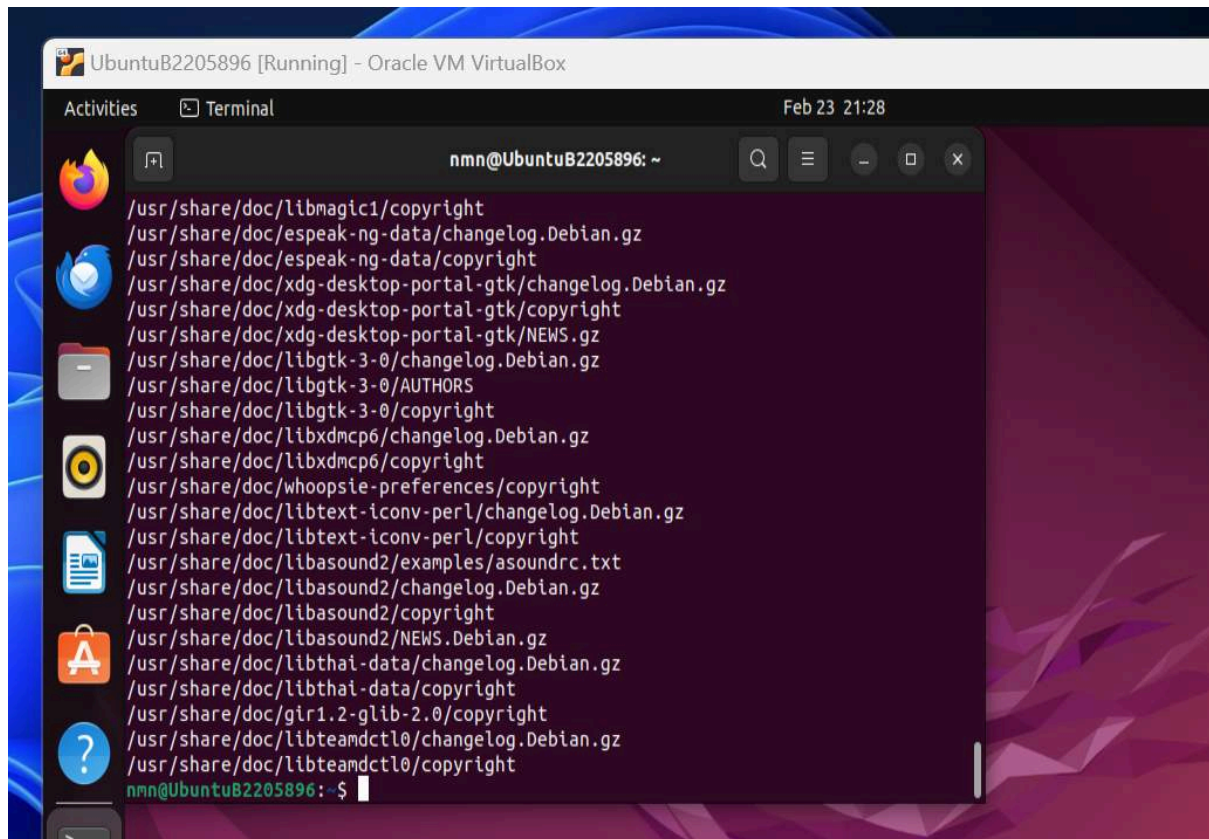
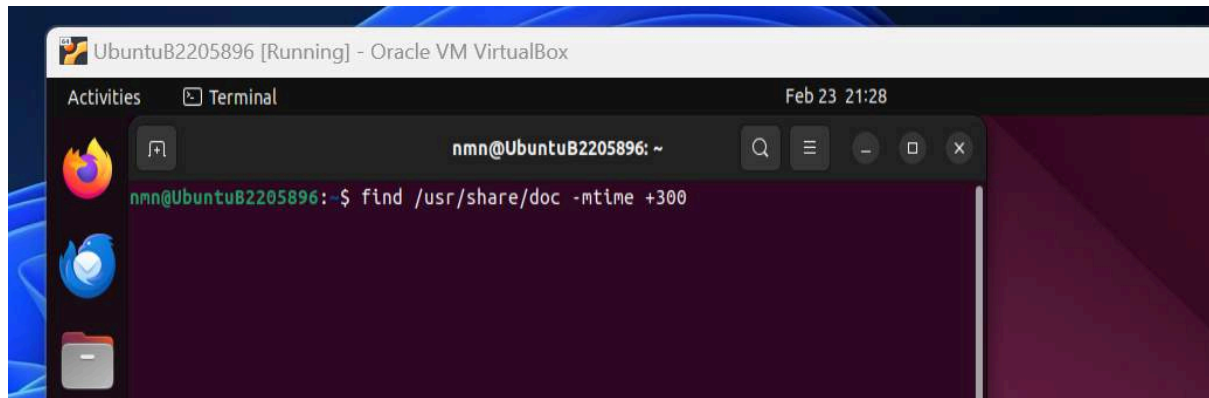
A terminal window titled 'UbuntuB2205896 [Running] - Oracle VM VirtualBox' showing the same user 'nmn' performing the same sequence of commands as in the first screenshot, but from the home directory. The 'find' command is repeated, showing the full path to the files in the home directory.

```
nmn@UbuntuB2205896:~$ mkdir TEST
nmn@UbuntuB2205896:~$ cd TEST
nmn@UbuntuB2205896:~/TEST$ touch{one,two,three}
touchone: command not found
nmn@UbuntuB2205896:~/TEST$ touch {one,two,three}
nmn@UbuntuB2205896:~/TEST$ cd
nmn@UbuntuB2205896:~$ chmod 777 $HOME/TEST/{one,two,three}
nmn@UbuntuB2205896:~$ find -perm -002 -type f -ls
1311242 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 ./TEST/two
1311247 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 ./TEST/three
1311190 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 ./TEST/one
nmn@UbuntuB2205896:~$ find $HOME -perm -002 -type f -ls
1311242 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 /home/nmn/TEST/two
1311247 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 /home/nmn/TEST/three
1311190 0 -rwxrwxrwx 1 nmn nmn 0 Feb 23 21:24 /home/nmn/TEST/one
nmn@UbuntuB2205896:~$
```

Question 6: Find files under the /usr/share/doc directory that have not been modified in more than 300 days.

Answer:

```
$ find /usr/share/doc -mtime +300
```



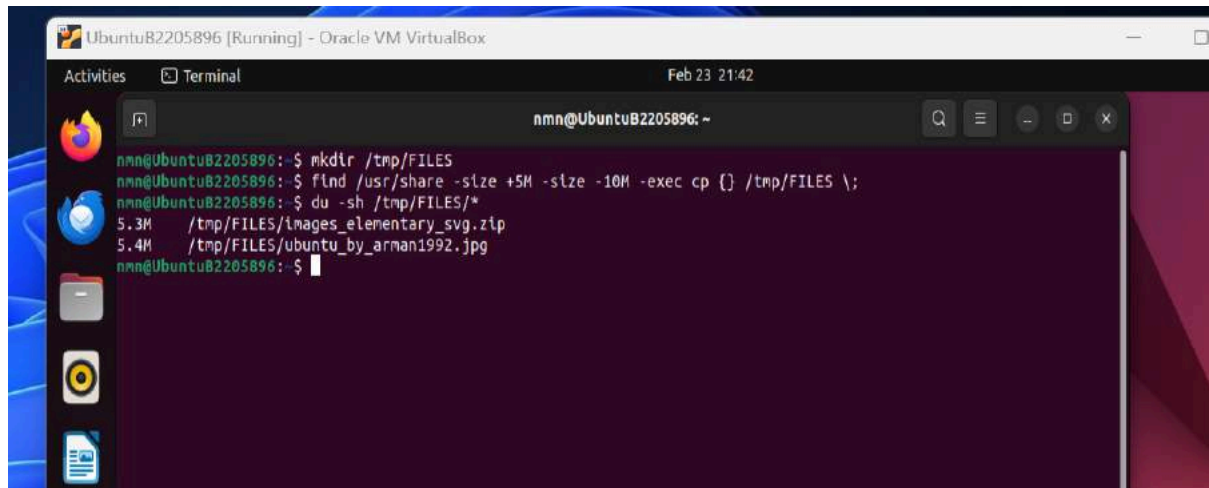
Question 7: Create a /tmp/FILES directory. Find all files under the /usr/share directory that are more than 5MB and less than 10MB and copy them to the /tmp/FILES directory.

Answer:

```
$ mkdir /tmp/FILES
```

```
$ find /usr/share -size +5M -size -10M -exec cp {} /tmp/FILES \;
```

```
$ du -sh /tmp/FILES/*
```

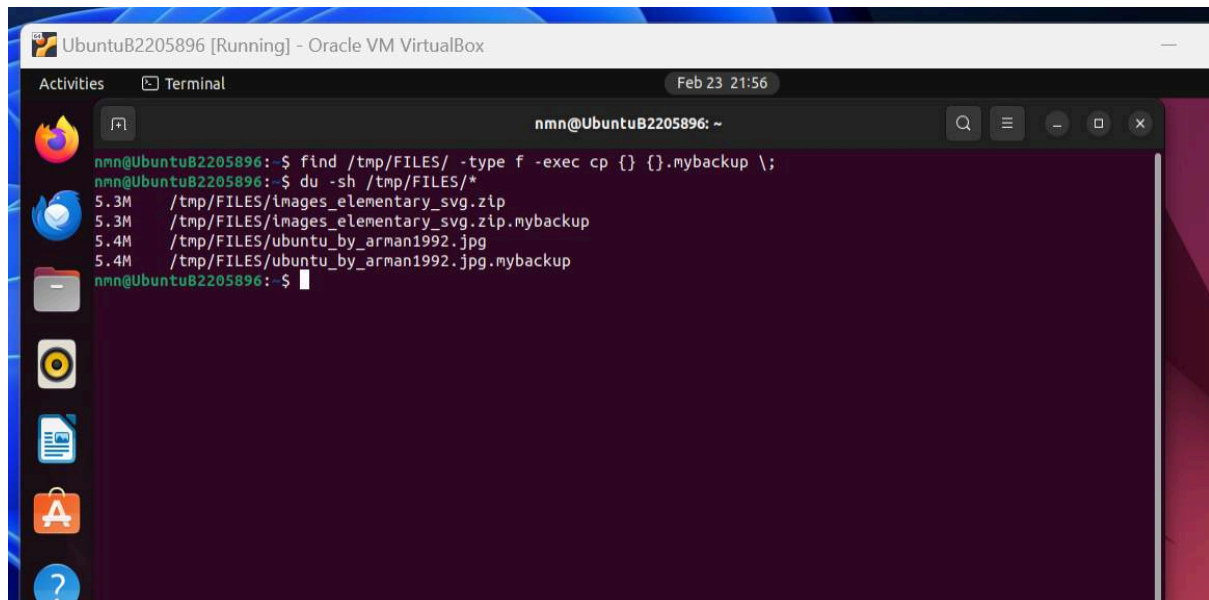



```
nmn@UbuntuB2205896: ~  
nmn@UbuntuB2205896:~$ mkdir /tmp/FILES  
nmn@UbuntuB2205896:~$ find /usr/share -size +5M -size -10M -exec cp {} /tmp/FILES \  
nmn@UbuntuB2205896:~$ du -sh /tmp/FILES/*  
5.3M    /tmp/FILES/images_elementary_svg.zip  
5.4M    /tmp/FILES/ubuntu_by_arman1992.jpg  
nmn@UbuntuB2205896:~$
```

Question 8: Find every file in the /tmp/FILES directory, and make a backup copy of each file in the same directory. Use each file's existing name, and just append .mybackup to create each backup file.

Answer:

\$ find /tmp/FILES/ -type f -exec cp {} {}.mybackup \;



```
nmn@UbuntuB2205896: ~  
nmn@UbuntuB2205896:~$ find /tmp/FILES/ -type f -exec cp {} {}.mybackup \  
nmn@UbuntuB2205896:~$ du -sh /tmp/FILES/*  
5.3M    /tmp/FILES/images_elementary_svg.zip  
5.3M    /tmp/FILES/images_elementary_svg.zip.mybackup  
5.4M    /tmp/FILES/ubuntu_by_arman1992.jpg  
5.4M    /tmp/FILES/ubuntu_by_arman1992.jpg.mybackup  
nmn@UbuntuB2205896:~$
```

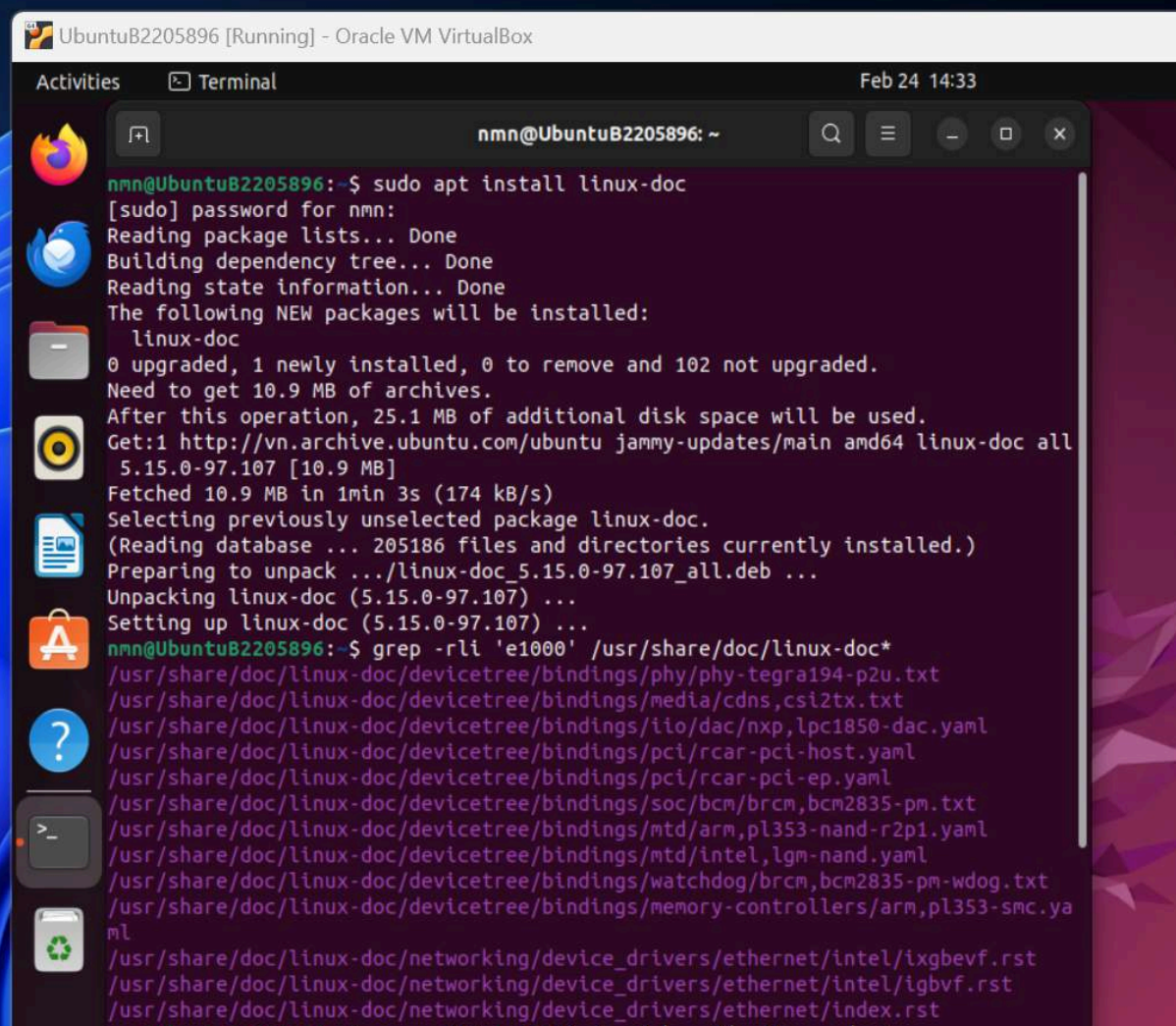
Question 9: Install the kernel-doc package in Fedora or Red Hat Enterprise Linux. Using grep, search inside the files contained in the /usr/share/doc/kernel-doc* directory for the term e1000 (case-insensitive) and list the names of the files that contain that term.

Answer:

\$ sudo apt install linux-doc

<Password>

\$ grep -rli 'e1000' /usr/share/doc/linux-doc*



```

nmn@UbuntuB2205896 [Running] - Oracle VM VirtualBox
Feb 24 14:33
nmn@UbuntuB2205896:~$ sudo apt install linux-doc
[sudo] password for nmn:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  linux-doc
0 upgraded, 1 newly installed, 0 to remove and 102 not upgraded.
Need to get 10.9 MB of archives.
After this operation, 25.1 MB of additional disk space will be used.
Get:1 http://vn.archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-doc all
5.15.0-97.107 [10.9 MB]
Fetched 10.9 MB in 1min 3s (174 kB/s)
Selecting previously unselected package linux-doc.
(Reading database ... 205186 files and directories currently installed.)
Preparing to unpack .../linux-doc_5.15.0-97.107_all.deb ...
Unpacking linux-doc (5.15.0-97.107) ...
Setting up linux-doc (5.15.0-97.107) ...
nmn@UbuntuB2205896:~$ grep -rli 'e1000' /usr/share/doc/linux-doc*
/usr/share/doc/linux-doc/devicetree/bindings/phy/phy-tegra194-p2u.txt
/usr/share/doc/linux-doc/devicetree/bindings/media/cdns,csi2tx.txt
/usr/share/doc/linux-doc/devicetree/bindings/iio/dac/nxp,lpc1850-dac.yaml
/usr/share/doc/linux-doc/devicetree/bindings/pci/rcar-pci-host.yaml
/usr/share/doc/linux-doc/devicetree/bindings/pci/rcar-pci-ep.yaml
/usr/share/doc/linux-doc/devicetree/bindings/soc/bcm/brcm,bcm2835-pm.txt
/usr/share/doc/linux-doc/devicetree/bindings/mtd/arm,pl353-nand-r2p1.yaml
/usr/share/doc/linux-doc/devicetree/bindings/mtd/intel,lgm-nand.yaml
/usr/share/doc/linux-doc/devicetree/bindings/watchdog/brcm,bcm2835-pm-wdog.txt
/usr/share/doc/linux-doc/devicetree/bindings/memory-controllers/arm,pl353-smc.yaml
/usr/share/doc/linux-doc/networking/device_drivers/ethernet/intel/ixgbevfv.rst
/usr/share/doc/linux-doc/networking/device_drivers/ethernet/intel/igbvf.rst
/usr/share/doc/linux-doc/networking/device_drivers/ethernet/index.rst

```

Question 10: Search for the e1000 term again in the same location, but this time list every line that contains the term and highlight the term in color

Answer:

```
$ cd /usr/share/doc/linux.doc*
```

```
$ grep -ri --color e1000
```

The screenshot shows a terminal window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The terminal is running a command to search for the string "e1000" in the directory "/usr/share/doc/linux-doc". The output lists various device tree bindings and their associated "e1000" identifiers.

```

nmn@UbuntuB2205896: /usr/share/doc/linux-doc
nmn@UbuntuB2205896:~$ cd /usr/share/doc/linux-doc*
nmn@UbuntuB2205896:/usr/share/doc/linux-doc$ grep -rt --color e1000
devicetree/bindings/phy/phy-tegra194-p2u.txt:p2u_hsio_0: phy@3e1000 {
devicetree/bindings/phy/phy-tegra194-p2u.txt: reg = <0x03e10000 0x10000>;
devicetree/bindings/media/cdns,csi2tx.txt:csi2tx: csi-bridge@0d0e1000 {
devicetree/bindings/media/cdns,csi2tx.txt: reg = <0x0d0e1000 0x1000>;
devicetree/bindings/lto/dac/nxp,lpc1850-dac.yaml: dac: dac@400e1000 {
devicetree/bindings/lto/dac/nxp,lpc1850-dac.yaml: reg = <0x400e1000 0x1000>;
devicetree/bindings/pcl/rcar-pcl-host.yaml: ranges = <0x01000000 0 0x00000000 0 0xf0e10000 0 0x00100000>,
devicetree/bindings/pcl/rcar-pcl-host.yaml: <0xf0e10000 0x100000>,
devicetree/bindings/soc/bcm/brcn,bcm2035-pm.txt: reg = <0x7e100000 0x114>,
devicetree/bindings/mtd/arm,pl353-nand-r2p1.yaml: ranges = <0x0 0x0 0xe1000000 0x1000000 /* NAND CS region */
devicetree/bindings/mtd/intel,lgm-nand.yaml: <0xe1000000 0x300>,
devicetree/bindings/watchdog/brcm,bcm2035-pm-wdog.txt: reg = <0x7e100000 0x28>;
devicetree/bindings/memory-controllers/arm,pl353-smc.yaml: ranges = <0x0 0x0 0xe1000000 0x1000000 /* NAND CS region */
networking/device_drivers/ethernet/intel/ixgbev.f.rst:https://sourceforge.net/projects/e1000
networking/device_drivers/ethernet/intel/ixgbev.f.rst:to e1000-devel@lists.sf.net.
networking/device_drivers/ethernet/intel/igbvf.rst:https://sourceforge.net/projects/e1000
networking/device_drivers/ethernet/intel/igbvf.rst:to e1000-devel@lists.sf.net.
networking/device_drivers/ethernet/index.rst: intel/e1000
networking/device_drivers/ethernet/index.rst: intel/e1000e
nmn@UbuntuB2205896:/usr/share/doc/linux-doc$

```


Chapter 7:

Question 1: Create a script in your \$HOME/bin directory called myownscript. When the script runs, it should output information that looks as follows:

Today is Sat Dec 10 15:45:04 EST 2016.

You are in /home/joe and your host is abc.example.com.

Of course, you need to read in your current date/time, current working directory, and hostname. Also, include comments about what the script does and indicate that the script should run with the /bin/bash shell.

Answer:

```
$ mkdir $HOME/bin
```

```
$ cd bin
```

```
$ touch myownscript
```

```
$ cd
```

```
$ chmod 755 $HOME/bin/myownscript
```

```
$ ./bin.myownscript
```

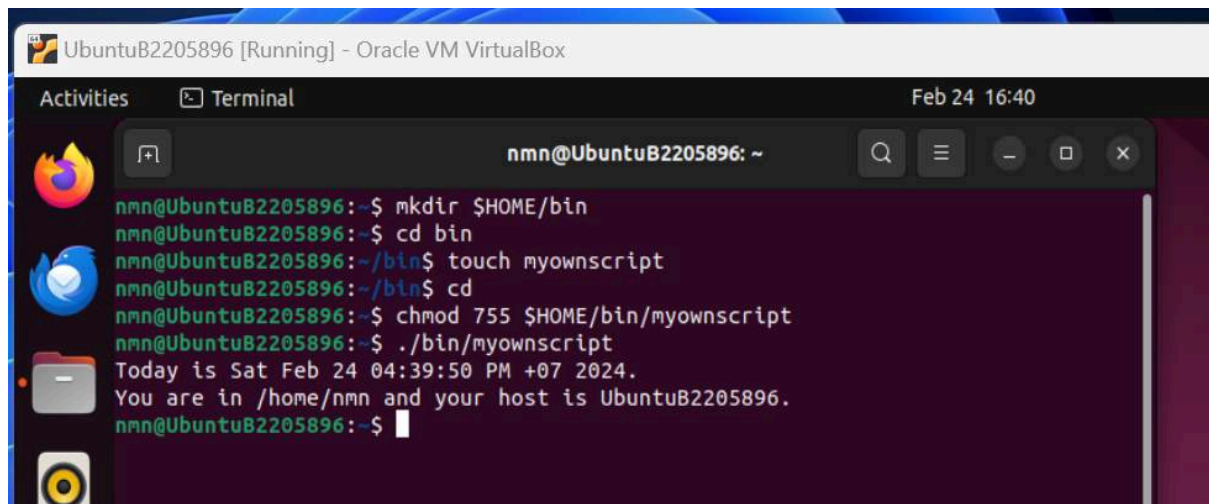
The script in myownscript file:

```
#!/bin/bash
```

```
# myownscript
```

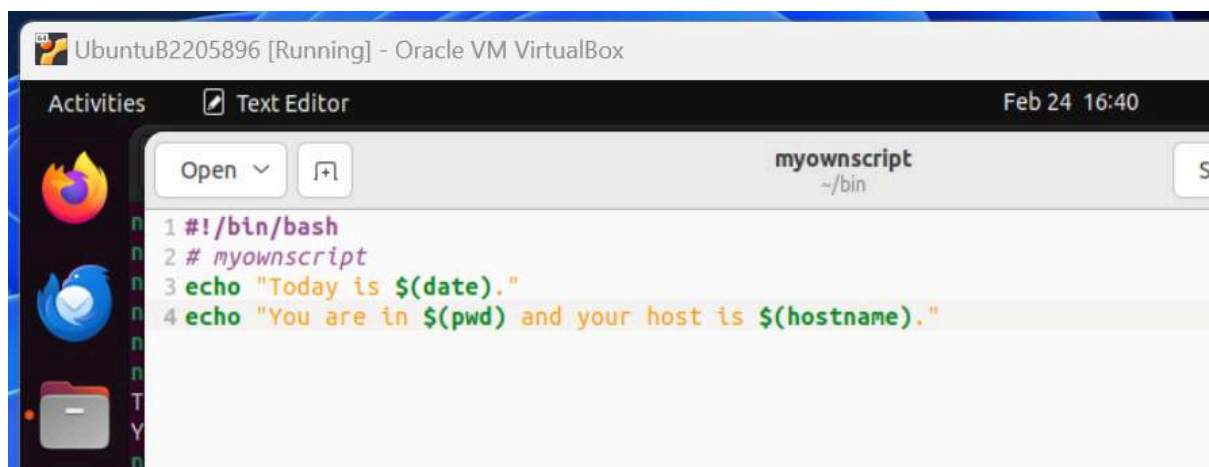
```
echo "Today is $(date)."
```

```
echo "You are in $(pwd) and your host is $(hostname)."
```



The screenshot shows a terminal window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The terminal output shows the following commands and their results:

```
nmn@UbuntuB2205896:~$ mkdir $HOME/bin
nmn@UbuntuB2205896:~$ cd bin
nmn@UbuntuB2205896:~/bin$ touch myownscript
nmn@UbuntuB2205896:~/bin$ cd
nmn@UbuntuB2205896:~$ chmod 755 $HOME/bin/myownscript
nmn@UbuntuB2205896:~$ ./bin/myownscript
Today is Sat Feb 24 04:39:50 PM +07 2024.
You are in /home/nmn and your host is UbuntuB2205896.
nmn@UbuntuB2205896:~$
```



The screenshot shows a text editor window titled "myownscript - /bin" in Oracle VM VirtualBox. The text editor displays the following content:

```
1 #!/bin/bash
2 # myownscript
3 echo "Today is $(date)."
```

Question 2: Create a script that reads in three positional parameters from the command line, assigns those parameters to variables named ONE, TWO, and THREE, respectively, and outputs that information in the following format:

There are X parameters that include Y.

The first is A, the second is B, the third is C.

Replace X with the number of parameters and Y with all parameters entered. Then replace A with the contents of variable ONE, B with variable TWO, and C with variable THREE.

Answer:

```
$ touch $HOME/bin/parameters
```

```
$ chmod 755 $HOME/bin/parameters
```

create a script contain:

```
#!/bin/bash
```

```
# parameters
```

```
ONE=$1
```

```
TWO=$2
```

```
THREE=$3
```

```
echo "There are $# parameters that include: $@"
```

```
echo "The first is $ONE, the second is $TWO, the third is $THREE."
```

The script in parameters:

```
#!/bin/bash
```

```
#parameters
```

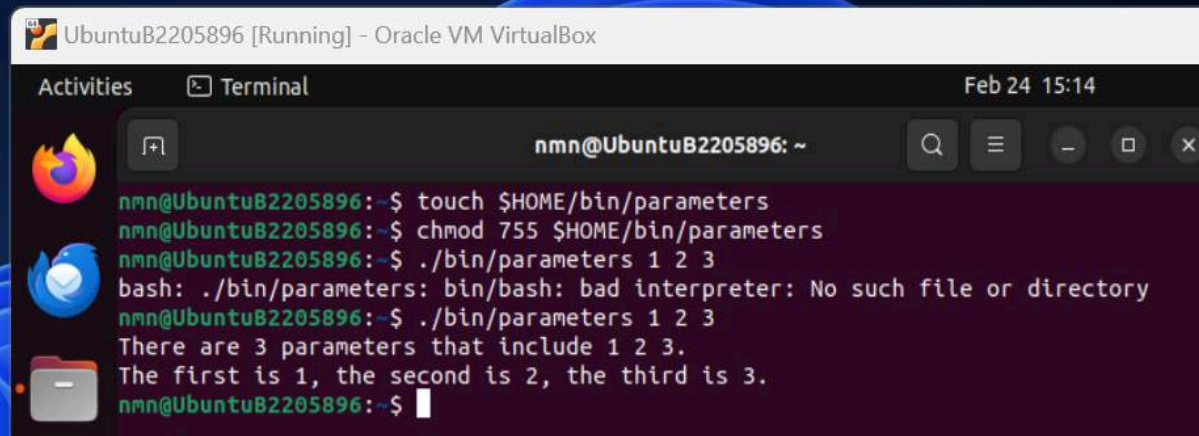
```
ONE=$1
```

```
TWO=$2
```

```
THREE=$3
```

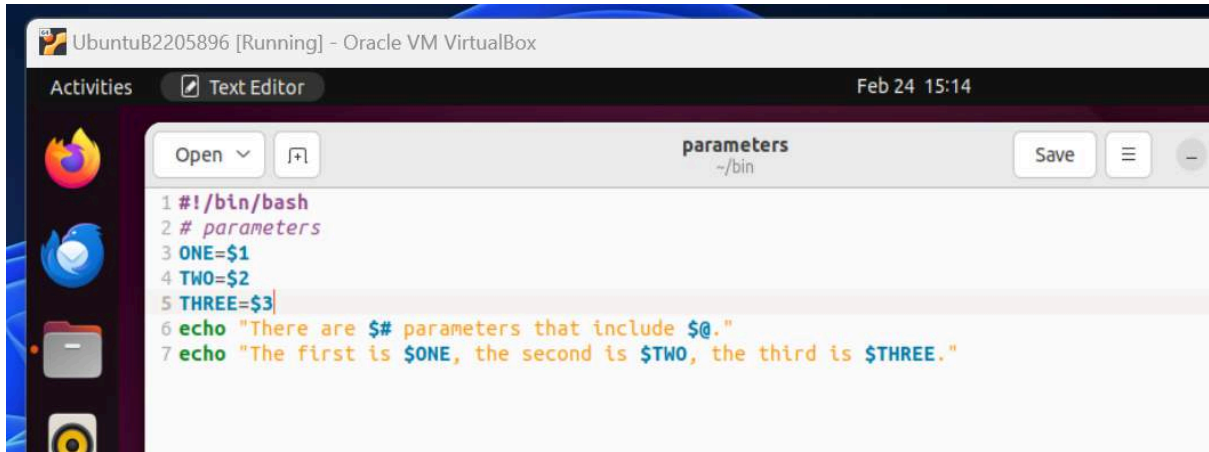
```
echo "There are $# parameters that include $@"
```

```
echo "The first is $ONE, the second is $TWO, the third is $THREE."
```



The screenshot shows a terminal window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
nmn@UbuntuB2205896:~$ touch $HOME/bin/parameters
nmn@UbuntuB2205896:~$ chmod 755 $HOME/bin/parameters
nmn@UbuntuB2205896:~$ ./bin/parameters 1 2 3
bash: ./bin/parameters: bin/bash: bad interpreter: No such file or directory
nmn@UbuntuB2205896:~$ ./bin/parameters 1 2 3
There are 3 parameters that include 1 2 3.
The first is 1, the second is 2, the third is 3.
nmn@UbuntuB2205896:~$
```

A screenshot of a text editor window titled 'parameters' in a virtual machine environment. The window shows a shell script with seven lines of code. The first line is a shebang, the second is a comment, and the next three are variable assignments. The last two lines are echo statements that use the variables and the number of command-line arguments. The window has a sidebar with application icons and a top bar with window controls and a date/time display.

```
1 #!/bin/bash
2 # parameters
3 ONE=$1
4 TWO=$2
5 THREE=$3
6 echo "There are $# parameters that include $@"
7 echo "The first is $ONE, the second is $TWO, the third is $THREE."
```

Question 3: Create a script that prompts users for the name of the street and town where they grew up. Assign town and street to variables called mytown and mystreet, and output them with a sentence that reads as shown in the following code (of course, \$mystreet and \$mytown will appear with the actual town and street the user enters):

The street I grew up on was \$mystreet and the town was \$mytown.

Answer:

```
$ touch $HOME/bin/myhome
```

```
$ chmod 755 $HOME/bin/myhome
```

```
$ ./bin/myhome
```

```
Duong926<Enter>
```

```
CanTho<Enter>
```

The script in myhome:

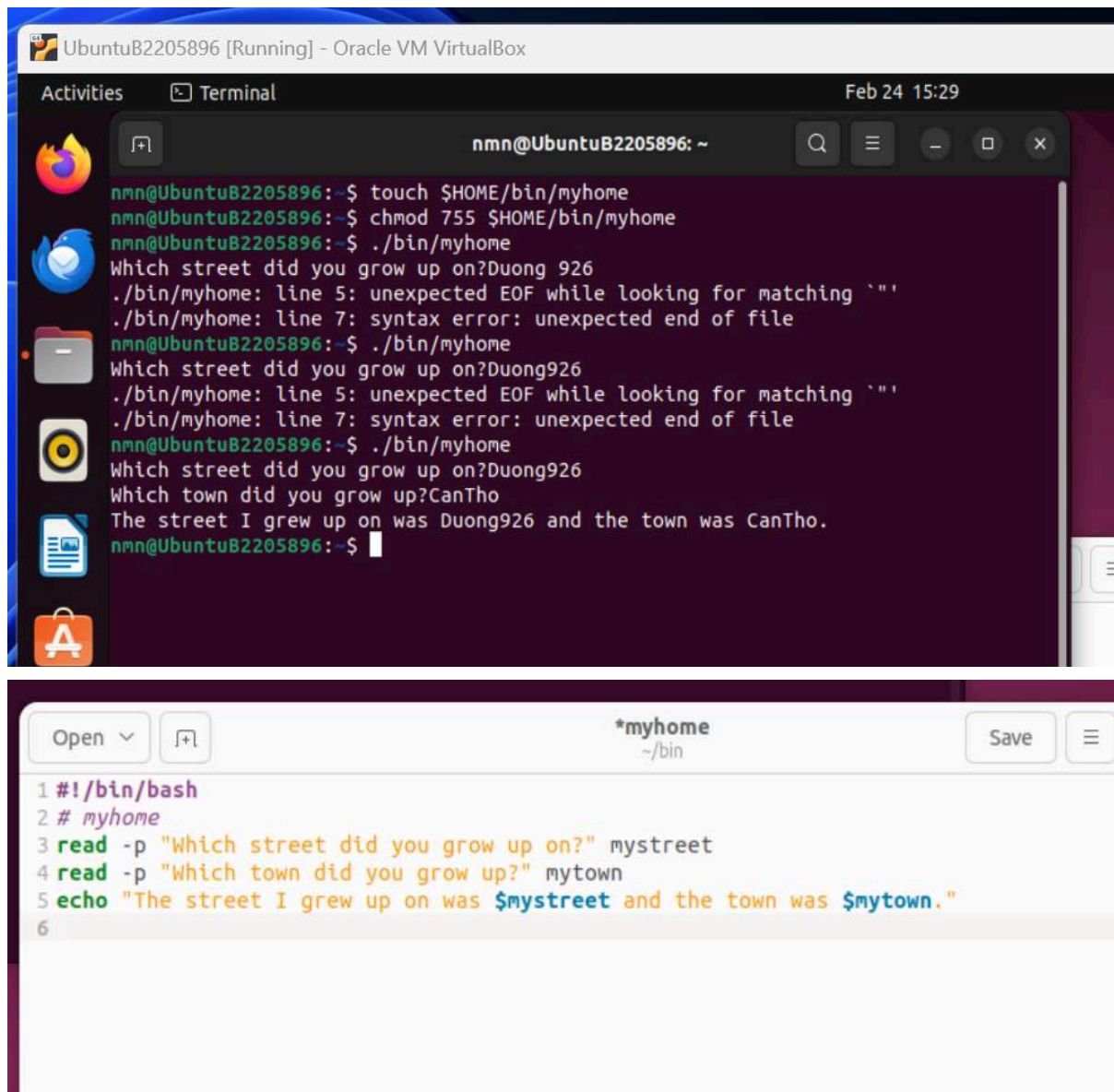
```
#!/bin/bash
```

```
# myhome
```

```
read -p "Which street did you grow up on?" mystreet
```

```
read -p "Which town did you grow up in?" mytown
```

```
echo "The street I grew up in was $mystreet and the town was $mytown."
```



```

nmn@UbuntuB2205896:~$ touch $HOME/bin/myhome
nmn@UbuntuB2205896:~$ chmod 755 $HOME/bin/myhome
nmn@UbuntuB2205896:~$ ./bin/myhome
Which street did you grow up on?Duong 926
./bin/myhome: line 5: unexpected EOF while looking for matching `"'
./bin/myhome: line 7: syntax error: unexpected end of file
nmn@UbuntuB2205896:~$ ./bin/myhome
Which street did you grow up on?Duong926
./bin/myhome: line 5: unexpected EOF while looking for matching `"'
./bin/myhome: line 7: syntax error: unexpected end of file
nmn@UbuntuB2205896:~$ ./bin/myhome
Which street did you grow up on?Duong926
Which town did you grow up?CanTho
The street I grew up on was Duong926 and the town was CanTho.
nmn@UbuntuB2205896:~$

```

```

1 #!/bin/bash
2 # myhome
3 read -p "Which street did you grow up on?" mystreet
4 read -p "Which town did you grow up?" mytown
5 echo "The street I grew up on was $mystreet and the town was $mytown."
6

```

Question 4: Create a script called myos that asks the user, “What is your favorite operating system?” Output an insulting sentence if the user types Windows or Mac. Respond “Great choice!” if the user types Linux. For anything else, say “Is <what is typed in> an operating system?”

Answer:

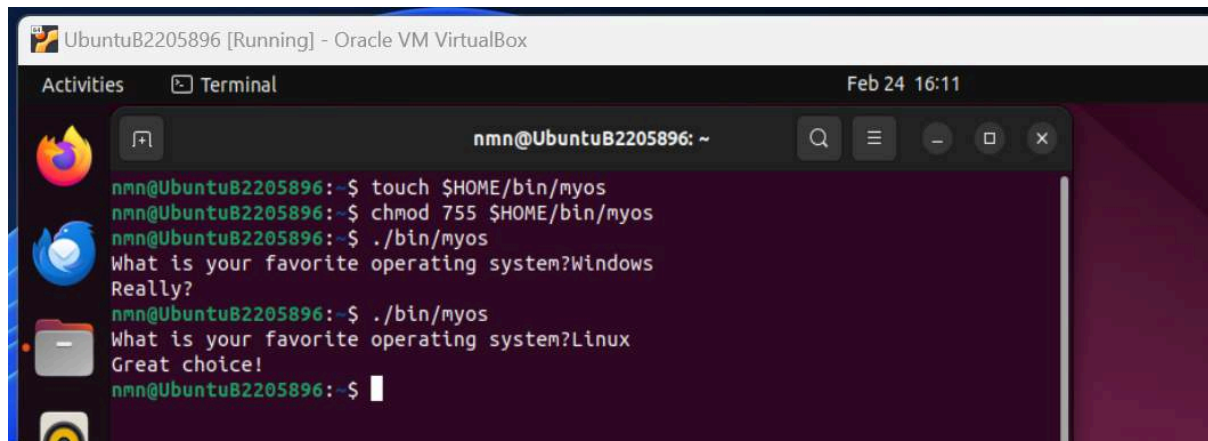
```

$ touch $HOME/bin/myos
$ chmod 755 $HOME/bin/myos
$ ./bin/myos
the script in myos file
#!/bin/bash
# myos
read -p “What is your favorite operating system?” input
if [ $input = Windows ] ; then
    echo “Really?”
elif [ $input = Mac ] ; then
    echo “Stupid”
elif [ $input = Linux ] ;then
    echo “ Great choice!”

```

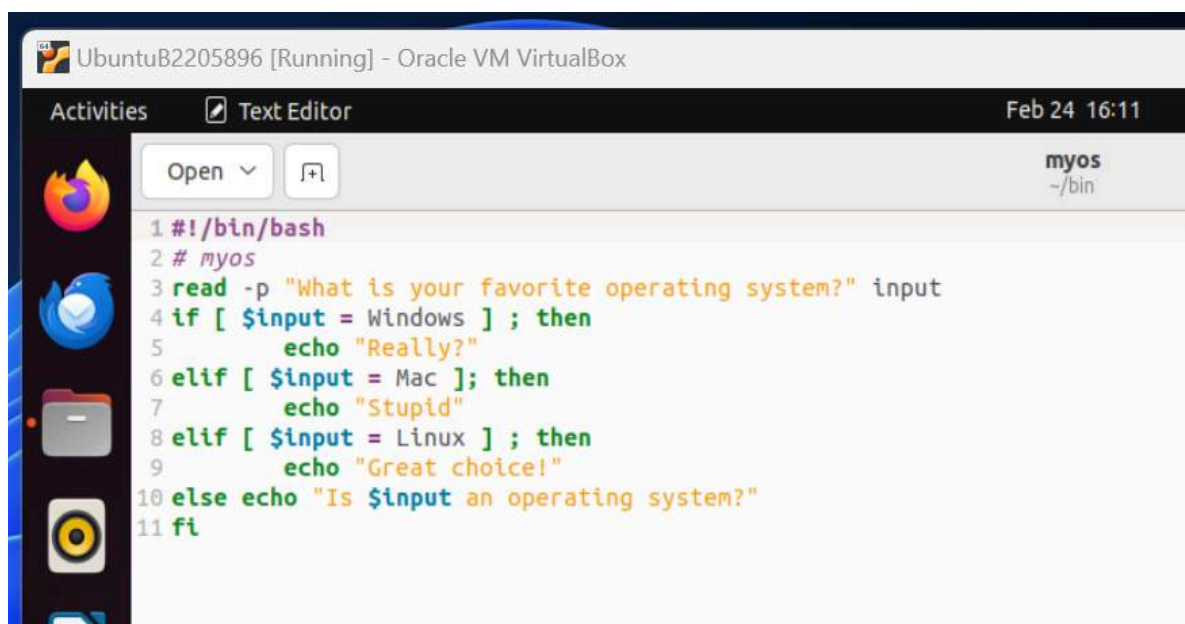


```
else echo "Is $input an operating system?"
fi
```



The screenshot shows a terminal window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The user is logged in as nmnn. The terminal shows the following commands and output:

```
nmnn@UbuntuB2205896:~$ touch $HOME/bin/myos
nmnn@UbuntuB2205896:~$ chmod 755 $HOME/bin/myos
nmnn@UbuntuB2205896:~$ ./bin/myos
What is your favorite operating system?Windows
Really?
nmnn@UbuntuB2205896:~$ ./bin/myos
What is your favorite operating system?Linux
Great choice!
nmnn@UbuntuB2205896:~$
```



The screenshot shows a text editor window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox". The file being edited is named "myos" and is located in the directory "~/bin". The code is as follows:

```
1 #!/bin/bash
2 # myos
3 read -p "What is your favorite operating system?" input
4 if [ $input = Windows ] ; then
5     echo "Really?"
6 elif [ $input = Mac ]; then
7     echo "Stupid"
8 elif [ $input = Linux ] ; then
9     echo "Great choice!"
10 else echo "Is $input an operating system?"
11 fi
```

Question 5: Create a script that runs the words moose, cow, goose, and sow through a for loop. Have each of those words appended to the end of the line "I have a...."

Answer:

```
$ touch $HOME/bin/forloop
```

```
$ chmod 755 $HOME/bin/forloop
```

```
$ ./bin/forloop
```

The script in forloop file:

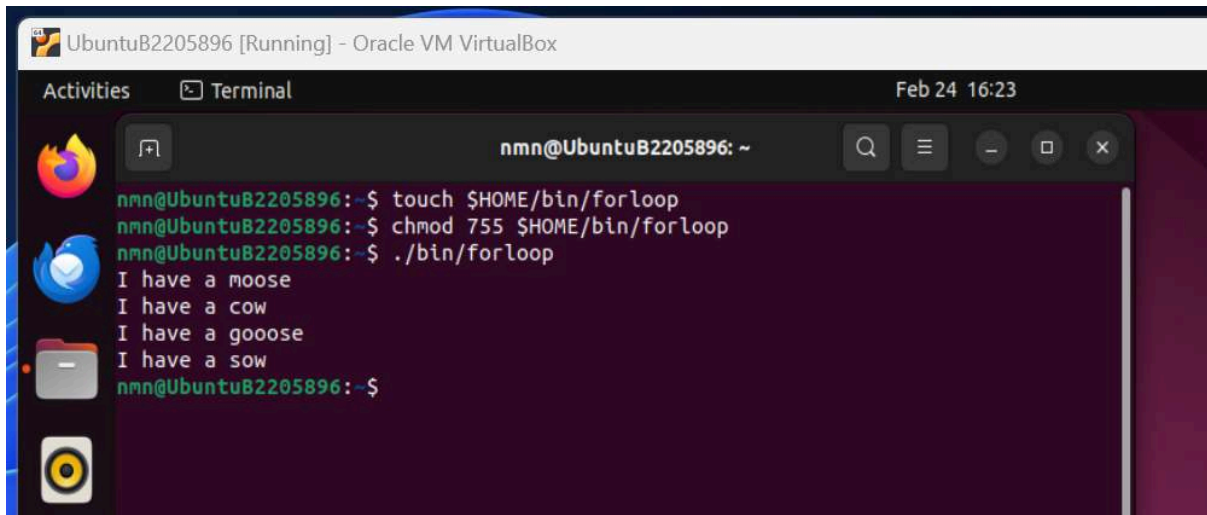
```
#!/bin/bash
```

```
#forloop
```

```
for forloop in moose cow goose sow ; do
```

```
    echo "I have a $forloop"
```

```
done
```



A terminal window titled "UbuntuB2205896 [Running] - Oracle VM VirtualBox" with a date and time of "Feb 24 16:23". The window shows a user named "nmn" at the prompt "nmn@UbuntuB2205896: ~". The user enters the following commands:

```
nmn@UbuntuB2205896:~$ touch $HOME/bin/forloop
nmn@UbuntuB2205896:~$ chmod 755 $HOME/bin/forloop
nmn@UbuntuB2205896:~$ ./bin/forloop
```

The output of the script is:

```
I have a moose
I have a cow
I have a gooose
I have a sow
nmn@UbuntuB2205896:~$
```



A text editor window titled "forloop" with a subtitle "~bin" and a date and time of "Feb 24 16:23". The window shows the following content:

```
1 #!/bin/bash
2 # forloop
3 for forloop in moose cow gooose sow ; do
4     echo "I have a $forloop"
5 done
```